(ii) Let $I = \int \frac{e^x}{e^x - 1} dx.$

Putting $e^x - 1 = u$, then $e^x dx = du$

Thus, $I = \int \frac{du}{u} = \log|u| + c = \log|e^x - 1| + c$

Therefore, $\int \frac{e^x}{e^x - 1} dx = \log|e^x - 1| + c.$

(iii) Let $I = \int \frac{1}{x \log x} dx.$

Putting $\log x = u$, then $\frac{1}{x} dx = du$

Thus, $I = \int \frac{du}{u} = \log|u| + c = \log|\log x| + c$

Therefore, $\int \frac{1}{x \log x} dx = \log|\log x| + c.$

(iv) Let $I = \int \frac{\sin x + \cos x}{\sin x - \cos x} dx.$

Putting $\sin x - \cos x = u$, then $(\cos x + \sin x)\, dx = du$

Thus, $I = \int \frac{du}{u} = \log|u| + c = \log|\sin x - \cos x| + c$

Therefore, $\int \frac{\sin x + \cos x}{\sin x - \cos x} dx = \log|\sin x - \cos x| + c$

(v) Let $I = \int \frac{\cos 2x}{(\sin x + \cos x)^2} dx = \int \frac{\cos 2x}{1 + \sin 2x} dx.$

Putting $1 + \sin 2x = u$, then $2 \cos 2x\, dx = du$

Thus, $I = \int \frac{du}{2u} = \frac{1}{2} \log|u| + c = \frac{1}{2} \log|1 + \sin 2x| + c.$

EXERCISE 11.6

Integrate the following with respect to x

(1) $\frac{x}{\sqrt{1+x^2}}$ (2) $\frac{x^2}{1+x^6}$ (3) $\frac{e^x - e^{-x}}{e^x + e^{-x}}$ (4) $\frac{10x^9 + 10^x \log_e 10}{10^x + x^{10}}$

(5) $\frac{\sin\sqrt{x}}{\sqrt{x}}$ (6) $\frac{\cot x}{\log(\sin x)}$ (7) $\frac{\operatorname{cosec} x}{\log\left(\tan\frac{x}{2}\right)}$ (8) $\frac{\sin 2x}{a^2 + b^2 \sin^2 x}$

(9) $\frac{\sin^{-1} x}{\sqrt{1-x^2}}$ (10) $\frac{\sqrt{x}}{1+\sqrt{x}}$ (11) $\frac{1}{x \log x \log(\log x)}$ (12) $\alpha\beta x^{\alpha-1} e^{-\beta x^\alpha}$

(13) $\tan x \sqrt{\sec x}$ (14) $x(1-x)^{17}$ (15) $\sin^5 x \cos^3 x$ (16) $\frac{\cos x}{\cos(x-a)}$

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